

Selim Sabag Romero Gonzalez

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Education

- **Doctor of Philosophy (Ph.D.)** Computational Science – The University of Texas at El Paso
 - Dissertation: ADVANCES IN ONE-ELECTRON SELF-INTERACTION-CORRECTION METHODS FOR ACCURATE AND EFFICIENT SELF-INTERACTION-FREE DENSITY FUNCTIONAL CALCULATIONS
- **Master of Science (M.S.)** – Computational Science – The University of Texas at El Paso
- **Master of Science (M.S.)** – Physics – The University of Texas at El Paso
- **Bachelor of Science – (B.S.)** Engineering Physics – Universidad Autónoma de Ciudad Juárez

Summary of Research Interests

My research integrates Foundational Quantum Computing with Electrical and Computer Engineering (ECE) applications in complex data analytics and simulation. Drawing on a strong background in theoretical physics and HPC, I specialize in developing and implementing both gate-based quantum algorithms (e.g., Qiskit) and Quantum Annealing (QA) approaches for solving complex, high-dimensional problems. Current work focuses on building novel quantum algorithms (including QA for enhanced feature selection) for classification in single-cell RNA sequencing (scRNA-seq) data, directly addressing challenges in cell differentiation and cancer drug delivery. My goal is to conduct foundational research in the experimental realization of quantum devices and interface circuits, leading to new systems for secure communications and integrated quantum-classical high-performance computing. My core goal is to establish scalable quantum-classical frameworks where quantum computing provides a high-dimensional, probabilistic kernel to accelerate complex data processing and optimize results, enabling solutions for complex and accurate representations of data-oriented problems with integrated quantum-classical high-performance computing.

Experience

Postdoctoral Research Associate Experience (10/2023-Present)

Single-Cell Data Core (SCDC), Texas A&M University

- Focuses on developing both classical and quantum algorithms to advance data-driven solutions in life sciences and cancer biology.
- Leveraging single-cell RNA sequencing and quantum computing for cancer research.
- Conducted in-depth analysis of single-cell data generated in collaboration with Dr. Chapkin's lab to illuminate the influence of malignant cells and pinpoint genes crucial for their propagation.
- Decrypted the key transition factors and signaling pathways underpinning their abnormal behavior, providing valuable insights into cancer progression.
- Currently developing novel quantum algorithms for accurate classification of malignant cells within single-cell RNA sequencing data, an innovative approach with the potential to revolutionize cancer diagnostics and personalized treatment strategies.
- Aggie Research Corps team leader of project (Fall 2025): "Machine Learning & Quantum Approaches to Deciphering Gene Signatures"

Graduate research Computational Science (2018-2023)

The University of Texas at El Paso

- 5 years of experience doing Fortran90 code development focused on density functional theory methods such as Perdew and Zunger self-interaction correction (PZSIC) and locally scaled self-interaction correction (LSIC).
- 5 years of experience doing High Performance Computing (HPC) using a Fortran90 MPI code, including valence selective orbital scaling self-interaction correction (vSOSIC) where a speed up of 10-50% is achieved compared to the PZSIC method.
- Developed a numerical implementation that optimizes an energy approximation through forward finite difference, providing a stable solution, and parallelized the subroutine with MPI for speeding up numerical integrations, achieving a speed up of 80%.
- Helped with code patches and included some implementations on the FLOSIC code (an open-source DFT code).
- Automatized tasks with Python and bash scripts to gather and process results in an efficient manner.

Teaching and Research Appointments (2015-2023)

- Teaching Assistant Grader for Differential Equations (09/2021 - 10/2023, UTEP).
- RA appointed by UTEP CoS (Summer 2023, UTEP).
- Research Assistant under Dr. Rajendra Zope (06/2020 - 05/2021, UTEP).
- Teaching Assistant MARCS tutoring (2018-2019, UTEP).
- Teaching Assistant Basic physics' Workshop (2016-2018, UTEP).
- Lecturer (06/2015 - 07/2016, Universidad Autónoma de Ciudad Juárez).

Publications

- "A Quantum Generative Framework for Modeling Single-Cell Transcriptomes with Gene-Gene and Cell-Cell Interactions", **Selim Romero**, Vignesh Kumar, Robert S. Chapkin, James J. Cai, arXiv:2510.12776, <https://doi.org/10.48550/arXiv.2510.12776>
- "Quantum Annealing for Enhanced Feature Selection in Single-Cell RNA Sequencing Data Analysis", **Selim Romero**, Shreyan Gupta, Victoria Gatlin, Robert S. Chapkin, James J. Cai, arXiv:2408.08867, <https://doi.org/10.48550/arXiv.2408.08867>
- "FGF8 induces bone and joint regeneration at digit amputation wounds in neonate mice", Ling Yu, Mingquan Yan, Sarah M. Wolff, Joseph D. Knue, Hannah M. Smith, Connor P. Dolan, Ken Muneoka, **Selim Romero**, James J. Cai, Carissa Yun, Devon J. Boland, Regina Brunauer, Lindsay A. Dawson, Vol. 202, January 2026, 117663, <https://doi.org/10.1016/j.bone.2025.117663>
- "Expression of Prooncogenic Nuclear Receptor 4A (NR4A)-Regulated Genes β 1-Integrin and G9a Inhibited by Dual NR4A1/2 Ligands", Lei Zhang, Victoria Gatlin, Shreyan Gupta, Michael L. Salinas, **Selim Romero**, James J. Cai, Robert S. Chapkin, Stephen Safe, Int. J. Mol. Sci. 2025, 26(8), 3909, <https://doi.org/10.3390/ijm4s26083909>
- "Exploring cell-to-cell variability and functional insights through differentially variable gene analysis", Victoria Gatlin, Shreyan Gupta, **Selim Romero**, Robert S. Chapkin, James J. Cai, npj Systems Biology and Applications volume 11, Article number: 29 (2025), DOI: 10.1038/s41540-025-00507-z
- "Simplification of the Fermi–Löwdin Self-Interaction Correction Method for Efficient Self-Interaction-Free Density Functional Calculations", **Selim Romero**, Yoh Yamamoto, Tunna Baruah, Rajendra Zope, ChemPhysChem, <https://doi.org/10.1002/cphc.202500086>
- "Self-consistent implementation of locally scaled self-interaction-correction method", Yoh Yamamoto, Tunna Baruah, Po-Hao Chang, **Selim Romero**, Rajendra Zope, J. Chem. Phys. 158, 064114 (2023), <https://doi.org/10.1063/5.0130436>
- "Spin-state gaps and self-interaction-corrected density functional approximations: Octahedral Fe(II) complexes as case study", **Selim Romero**, Tunna Baruah and Rajendra Zope, J. Chem. Phys. 158, 054305 (2023), <https://doi.org/10.1063/5.0133999>
- "Local self-interaction correction method with a simple scaling factor", **Selim Romero**, Yoh Yamamoto, Tunna Baruah and Rajendra Zope, Phys. Chem. Chem. Phys., 2021, 23, 2406, DOI: <https://doi.org/10.1039/D0CP06282K>
- "A Step in the Direction of Resolving the Paradox of Perdew-Zunger Self-interaction Correction. II. Gauge Consistency of the Energy Density at Three Levels of Approximation", Puskar Bhattarai, Kamal Wagle, Chandra Shahi, Yoh Yamamoto, **Selim Romero**, Biswajit Santra, Rajendra R. Zope, Juan E. Peralta, Koblar A. Jackson, and John P. Perdew, J. Chem. Phys. 152, 214109 (2020); <https://doi.org/10.1063/5.0010375>
- "Improvements in the orbitalwise scaling down of Perdew–Zunger self-interaction correction in many-electron regions", Yoh Yamamoto, **Selim Romero**, Tunna Baruah and Rajendra R. Zope, J. Chem. Phys. 152, 174112 (2020), DOI: <https://doi.org/10.1063/5.0004738>
- "A GEANT4 Study of a Gamma-ray Collimation Array", J. A. Lopez, **S. S. Romero**, O. Hernandez Rodriguez, J. Holmes and R. Alarcon, Journal of Nuclear Physics, Material Sciences, Radiation and Applications, 7(2), 217-221., 2020, DOI: <https://doi.org/10.15415/jnp.2020.72028>

Peer review activity:

- Molecular Docking via Weighted Subgraph Isomorphism on Quantum Annealers, Journal: Quantum Science and Technology, Review Dates: September 8, 2025; August 15, 2025; June 29, 2025
- Towards Arbitrary QUBO Optimization: Analysis of Classical and Quantum-Activated Feedforward Neural Networks, Journal: Machine Learning: Science and Technology, Review Dates: March 30, 2025; December 29, 2024

Workshops and trainings:

- Qiskit Global Summer School 2025 - Quantum Excellence (Issued Aug 2025, IBM)
- IOP Trusted Reviewer status (Issued Feb 2025, IOP Publishing)
- Peer Review Excellence online training graduate (Issued Dec 2024, IOP Publishing)
- Qiskit Global Summer School 2024 - Quantum Excellence (Issued Aug 2024, IBM)
- Fundamentals of Accelerated Computing with CUDA C/C++ (Issued Sep 2024, NVIDIA)
- Qiskit Global Summer School 2023: Theory To Implementation (June 2023)
- NERSC Modern Fortran Basics (FUN training) (July 2023)
- Codee Training Series: Write accelerated GPU code at Expert Level (April 2023)
- N-Ways To GPU Programming Bootcamp and Beyond for NERSC (April 2023)
- Intel Developer Tools Optimization Training (Webinar) (June 2022)

Additional Skills

- Proficient Linux and Bash script
- Expert in HPC code for density functional theory implementations
- Expert in Fortran90 oriented to density functional development
- Competent in MPI/OpenMP parallelization for HPC computing
- Basic proficiency in CUDA Fortran programming
- Skilled in Matlab/Octave for numerical methods implementation
- Skilled in Python and R programming for data analysis
- Capable in C programming oriented to numerical methods
- Knowledgeable in Mathematica and Maxima
- Knowledgeable in using Density functional theory packages such as NRLMOL, PySCF, PSI4 and Gaussian
- Single-Cell data analysis with Scanpy, Signac and Seurat

Language Skills

- Spanish – Native Language
- English – Professional Proficiency

Presentations:

- TAMU Postdoctoral Symposium 2025: QUBO Feature selection on scRNA-seq presentations drug-discovery
- TAMU HPRC Symposium 2025: QUBO Feature selection on scRNA-seq presentations drug-discovery
- CPRIT TAMU Cancer Symposium 2025: QUBO Feature selection on scRNA-seq presentations drug-discovery
- ICIBM 2024: QUBO Feature selection on scRNA-seq presentations cell differentiation

- TAMU Postdoctoral Symposium 2024: QUBO Feature selection on scRNA-seq presentations cell differentiation
- CPRIT TAMU Cancer Symposium 2024: Exploring single-cell to cell variability
- ACS Fall meeting 2022: Talk "SCAN family of functionals in the FLOSIC code"
- FLOSIC meeting 2022: Talk "Selectively orbital-scaled self-interaction-correction for efficient FLOSIC calculations toward realistic systems"
- APS march meeting 2021, Talk : "Local scaled self-interaction-correction method using FermiLöwdin orbitals and a simple scaling factor"
- FLOSIC meeting 2020: Talk "Assessment of Local Scaling Self-Interaction correction with a simpler scaling factor"
- UTEP GradEXPO 2021: Poster "Performance of Local-Self Interaction Correction using the density ratio as a scaling factor"
- FLOSIC meeting 2019: Poster "Study of Local Scaling Self-interaction correction with a simpler scaling factor"
- APS Fall Texas section 2017: "Geant4 simulator for gamma collimator to obtain the requirements of a measuring device and to track them"